POWER SYSTEMS

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AUTOMATIC DAY / NIGHT MIRROR

DESCRIPTION

The automatic dimming inside day/night rear view mirror system is a completely self-contained unit that replaces the standard equipment inside rear view mirror. This system will automatically change the reflectance of the inside rear view mirror to protect the driver from the unwanted headlight glare of trailing vehicles while driving at night. The automatic day/night inside mirror receives ignition switched battery current through a fuse in the junction block, and will only operate when the ignition switch is in the On position.

Contained within the mirror is a compass/temperature display as well as two LED reading lamps.

The automatic day/night mirror sensitivity cannot be repaired or adjusted. If any component of this unit is inoperative or damaged, the entire automatic day/night inside rear view mirror unit must be replaced.

OPERATION

The automatic day/night mirror is equipped with three buttons: the left switch for the left LED lamp, the right switch for the right side LED lamp and the center switch for the compass/temperature function. Pressing the left and right switch simultaneously will turn the auto dim function on or off. A green light next to the right button will indicate when the dimming feature is activated. The mirror also senses the backup lamp circuit, and will automatically disable its self-dimming feature whenever the transmission gear selector is in the Reverse position.

The compass/temperature display provides the outside temperature and one of eight compass headings. Press and release the center button once within 3 seconds to display compass/temperature (Fahrenheit). Press and release the center button twice within 3 seconds to display compass/temperature (Celsius). Press and release the center button three times within 3 seconds to deactivate the feature.

A thin layer of electrochromatic material between two pieces of conductive glass make up the face of the mirror. Two photocell sensors are used to monitor light levels and adjust the reflectance of the mirror. The ambient photocell sensor faces forward, to detect the outside light levels. The headlamp sensor is located on the mirror housing just to the left of the switch and facing rearward, to detect the light level received at the rear window side of the mirror. When the difference between the two light levels becomes too great (the light level received at the rear of the mirror is much higher than that at the front of the mirror), the mirror begins to darken.

DIAGNOSIS AND TESTING - AUTOMATIC DAY / NIGHT MIRROR

For complete circuit diagrams, refer to the appropriate wiring information.

(1) Check for a blown fuse. If OK, go to Step 2. If not OK, repair the shorted circuit or component as required and replace the faulty fuse.

(2) Turn the ignition switch to the On position. Check for battery voltage at the fuse. If OK, go to Step 3. If not OK, repair the open circuit to the ignition switch as required.

(3) Turn the ignition switch to the Off position. Disconnect and isolate the battery negative cable. Unplug the wire harness connector from the automatic day/night mirror. Connect the battery negative cable. Turn the ignition switch to the On position. Check for battery voltage at the fused ignition switch output (run/start) circuit cavity of the automatic day/night mirror wire harness connector. If OK, go to Step 4. If not OK, repair the open circuit as required.

(4) Turn the ignition switch to the Off position. Disconnect and isolate the battery negative cable. Check for continuity between the ground circuit cavity of the automatic day/night mirror wire harness connector and a good ground. There should be continuity. If OK, go to Step 5. If not OK, repair the circuit to ground as required.

(5) Connect the battery negative cable. Turn the ignition switch to the On position. Set the parking brake. Place the transmission gear selector lever in